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Practical 7

Aim: Web Security with SSL/TLS

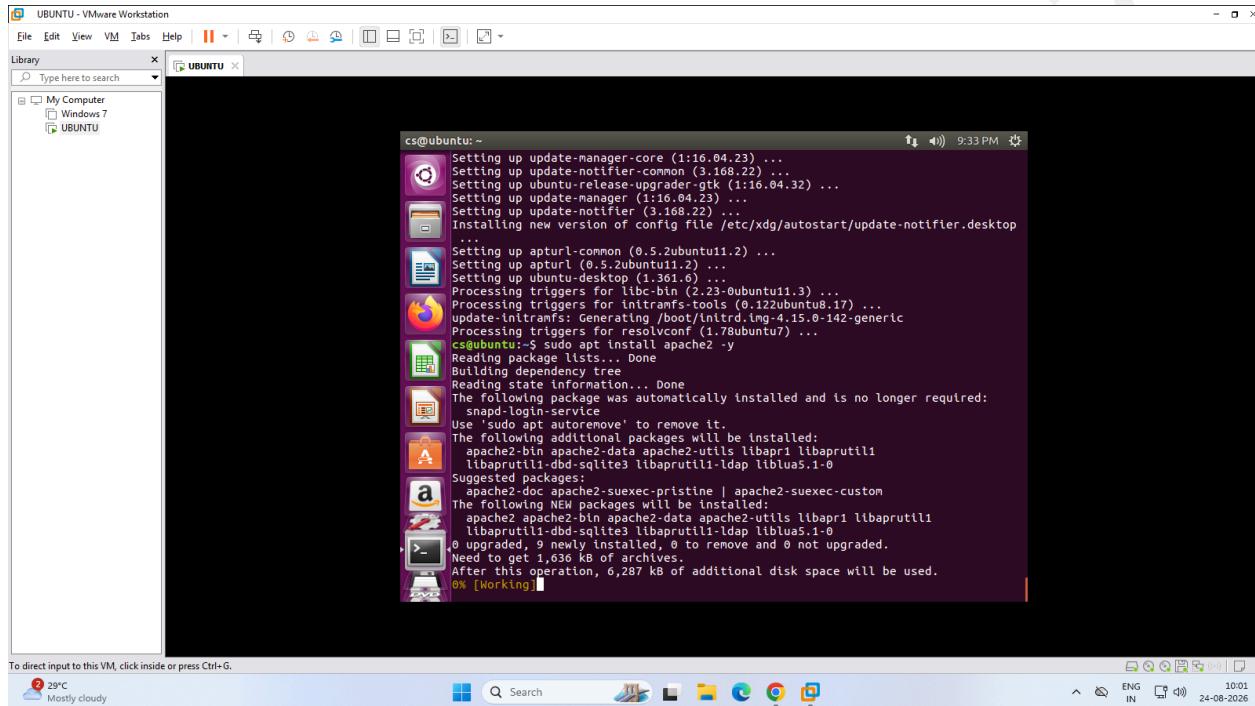
Step 1: Update the System Open Terminal and run:

```
sudo apt update
```

```
sudo apt upgrade -y
```

Step 2: Install Apache Web Server

```
sudo apt install apache2 -y
```

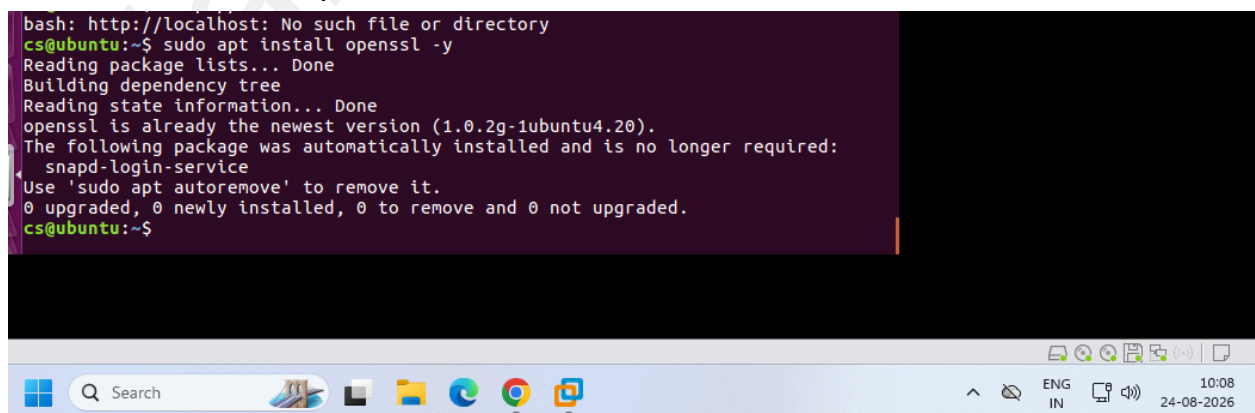


The screenshot shows a terminal window within a VMware Workstation environment. The terminal output displays the results of running 'sudo apt update' and 'sudo apt install apache2 -y'. The system is updating several packages, including 'update-manager-core', 'update-manager', 'update-notifier', 'apturl-common', 'apturl', and 'ubuntu-desktop'. After the update, the terminal shows the installation of 'apache2'. It lists additional packages to be installed, suggested packages, and the disk space requirements. The installation progress is shown as '0% Working'.

Step 3: Install OpenSSL OpenSSL is used to create and manage SSL/TLS certificates.

```
sudo apt install openssl -y
```

Check the version: openssl version



The screenshot shows a terminal window where the command 'sudo apt install openssl -y' has been executed. The output indicates that 'openssl' is already the newest version (1.0.2g-1ubuntu4.20). It also lists additional packages to be installed, suggested packages, and the disk space requirements. The installation progress is shown as '0% Working'.

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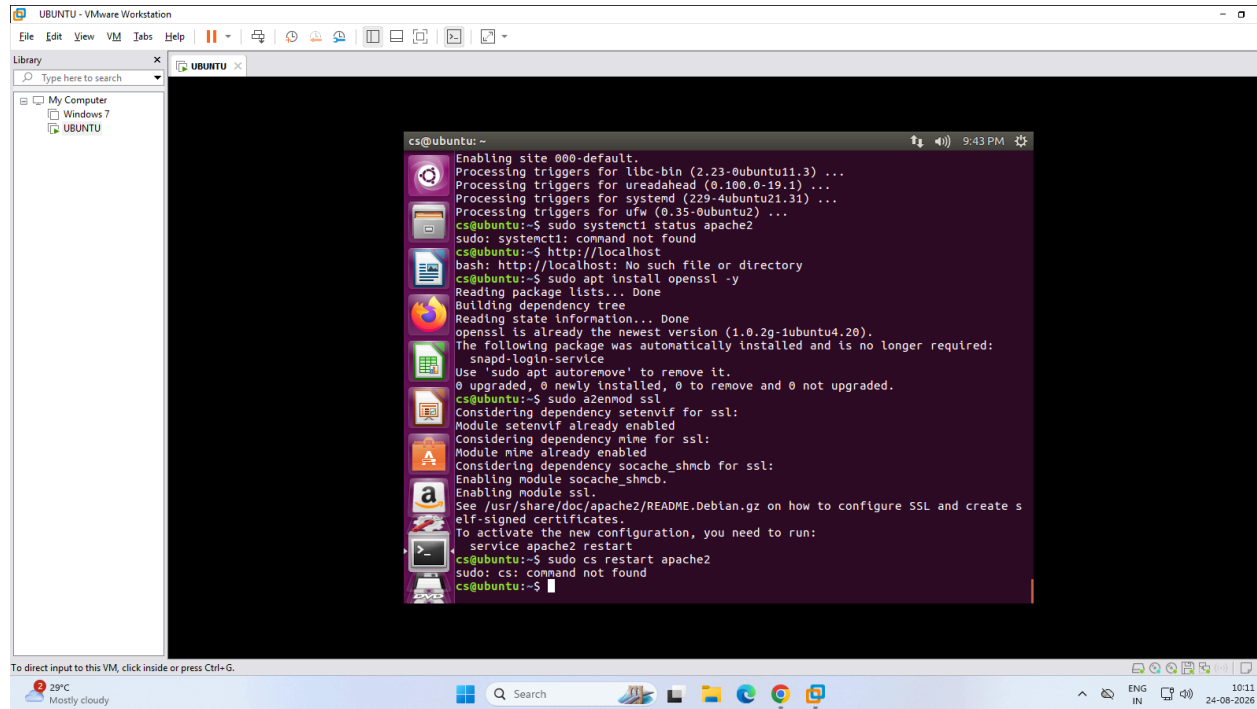
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Step 4: Enable SSL Module in Apache

Restart Apache:

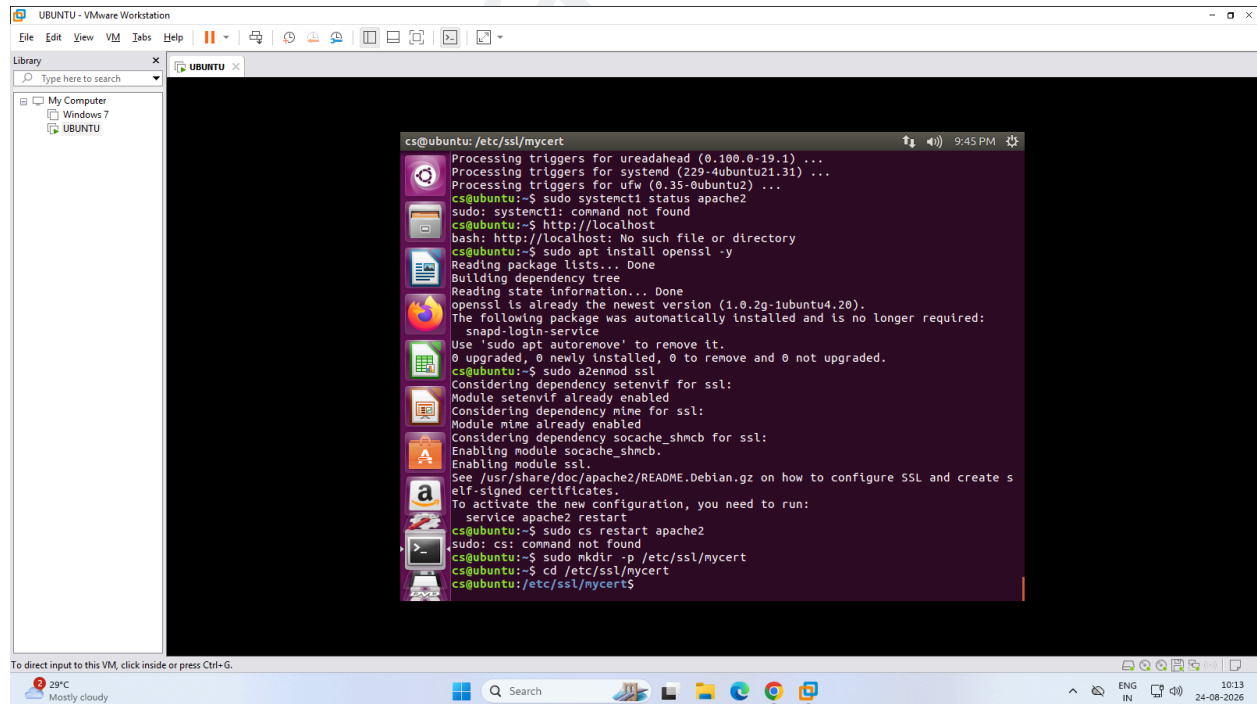
`sudo systemctl restart apache2`



```
cs@ubuntu:~$ sudo systemctl status apache2
sudo: systemctl: command not found
cs@ubuntu:~$ http://localhost
bash: http://localhost: No such file or directory
cs@ubuntu:~$ sudo apt install openssl -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
openssl is already the newest version (1.0.2g-1ubuntu4.20).
The following package was automatically installed and is no longer required:
  snapd-login-service
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
cs@ubuntu:~$ sudo a2enmod ssl
Considering dependency setenvif for ssl:
Module setenvif already enabled
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create s
elf-signed certificates.
To activate the new configuration, you need to run:
  service apache2 restart
cs@ubuntu:~$ sudo cs restart apache2
sudo: cs: command not found
cs@ubuntu:~$
```

Step 5: Create a Directory for Certificates

`sudo mkdir -p /etc/ssl/mycert`



```
cs@ubuntu:/etc/ssl/mycert$
Processing triggers for ureadahead (0.100.0-19.1) ...
Processing triggers for systemd (229-4ubuntu21.31) ...
Processing triggers for ufw (0.35-0ubuntu2) ...
cs@ubuntu:~$ sudo systemctl status apache2
sudo: systemctl: command not found
cs@ubuntu:~$ http://localhost
bash: http://localhost: No such file or directory
cs@ubuntu:~$ sudo apt install openssl -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
openssl is already the newest version (1.0.2g-1ubuntu4.20).
The following package was automatically installed and is no longer required:
  snapd-login-service
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
cs@ubuntu:~$ sudo a2enmod ssl
Considering dependency setenvif for ssl:
Module setenvif already enabled
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create s
elf-signed certificates.
To activate the new configuration, you need to run:
  service apache2 restart
cs@ubuntu:~$ sudo cs restart apache2
sudo: cs: command not found
cs@ubuntu:~$ sudo mkdir -p /etc/ssl/mycert
cs@ubuntu:~$ cd /etc/ssl/mycert
cs@ubuntu:/etc/ssl/mycert$
```

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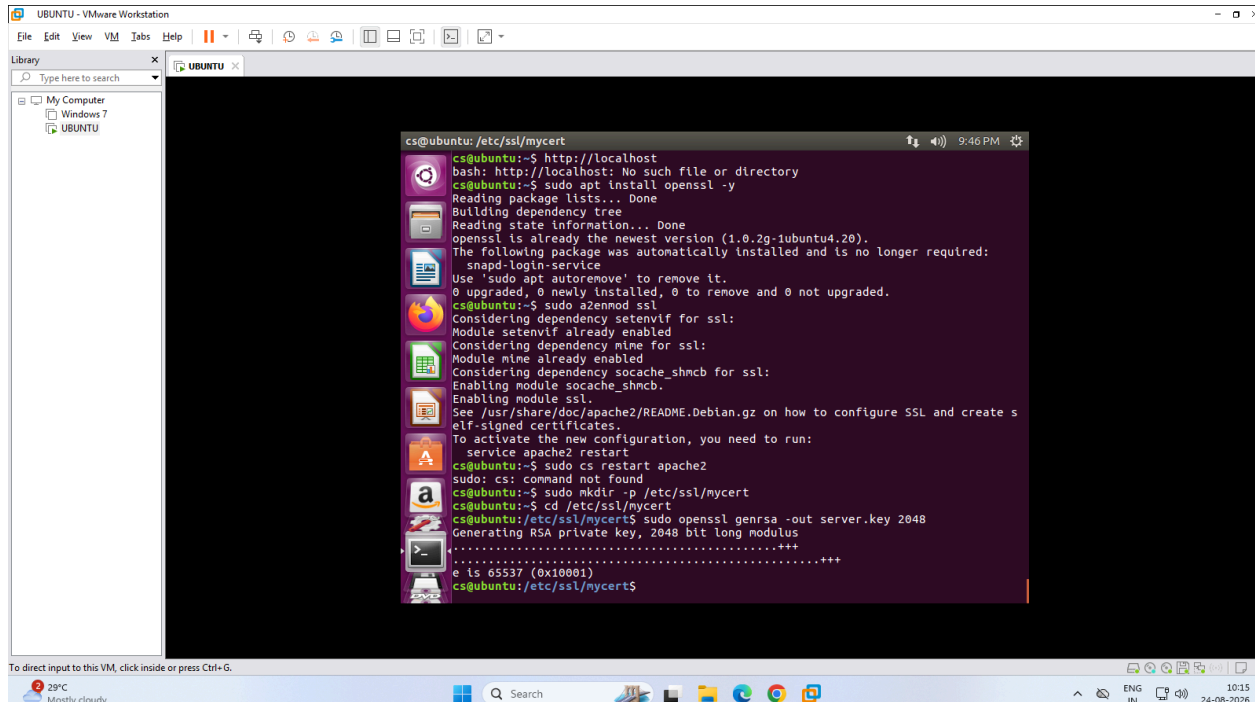
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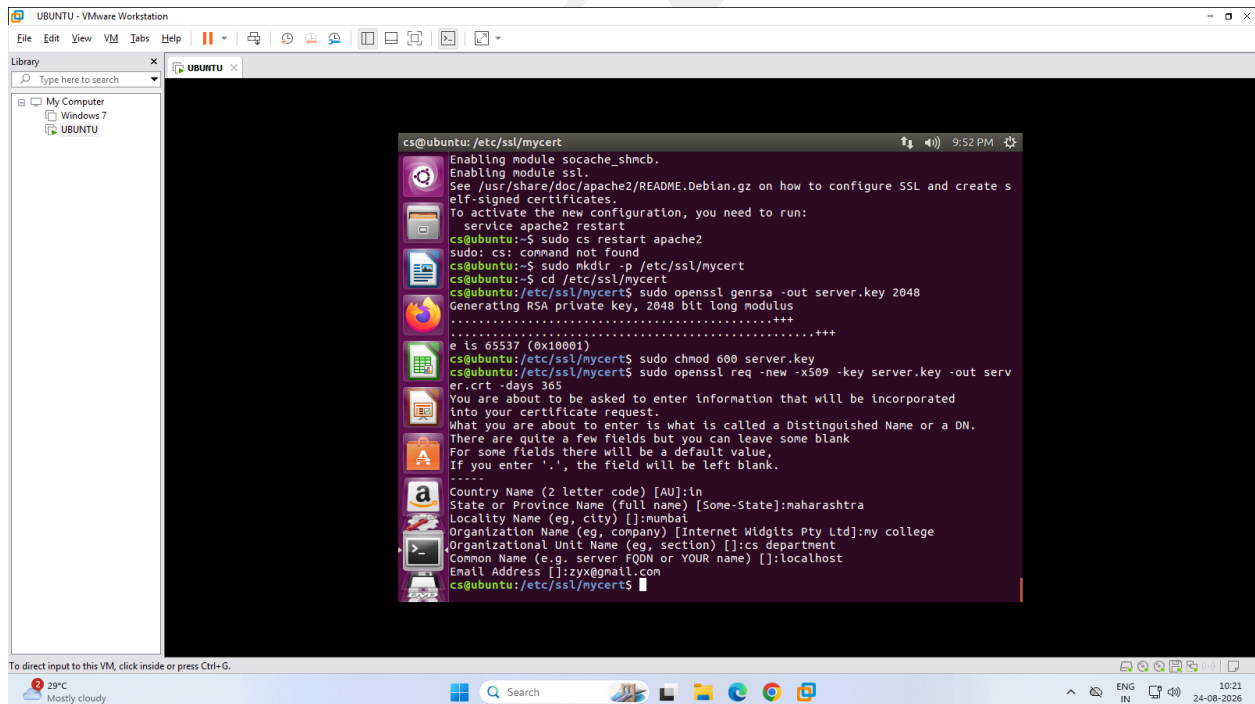
Step 6: Generate a Private Key



```
cs@ubuntu:~$ http://localhost
bash: http://localhost: No such file or directory
cs@ubuntu:~$ sudo apt install openssl -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
openssl is already the newest version (1.0.2g-1ubuntu4.20).
The following package was automatically installed and is no longer required:
  snapd-login-service
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
cs@ubuntu:~$ sudo a2enmod ssl
Considering dependency setenvif for ssl:
Module setenvif already enabled
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create s
elf-signed certificates.
To activate the new configuration, you need to run:
  service apache2 restart
cs@ubuntu:~$ sudo cs restart apache2
sudo: cs: command not found
cs@ubuntu:~$ sudo mkdir -p /etc/ssl/mycert
cs@ubuntu:~$ cd /etc/ssl/mycert
cs@ubuntu:~$ sudo openssl genrsa -out server.key 2048
Generating RSA private key, 2048 bit long modulus
.....+++
e is 65537 (0x10001)
cs@ubuntu:~$ cd /etc/ssl/mycert$
```

Step 7: Generate a Self-Signed SSL Certificate Run:

`sudo openssl req -new -x509 -key server.key -out server.crt -days 365`



```
cs@ubuntu:~$ sudo openssl req -new -x509 -key server.key -out server.crt -days 365
Generating RSA private key, 2048 bit long modulus
.....+++
e is 65537 (0x10001)
cs@ubuntu:~$ cd /etc/ssl/mycert$ sudo chmod 600 server.key
cs@ubuntu:~$ cd /etc/ssl/mycert$ sudo openssl req -new -x509 -key server.key -out serv
er.crt -days 365
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:in
State or Province Name (full name) [Some-State]:maharashtra
Locality Name (eg, city) []:mumbai
Organization Name (eg, company) [Internet Widgits Pty Ltd]:my college
Organizational Unit Name (eg, section) []:cs department
Common Name (e.g. server FQDN or YOUR name) []:localhost
Email Address []:zyx@gmail.com
cs@ubuntu:~$ cd /etc/ssl/mycert$
```

Step 8: Configure Apache for HTTPS

Enable the default SSL site:

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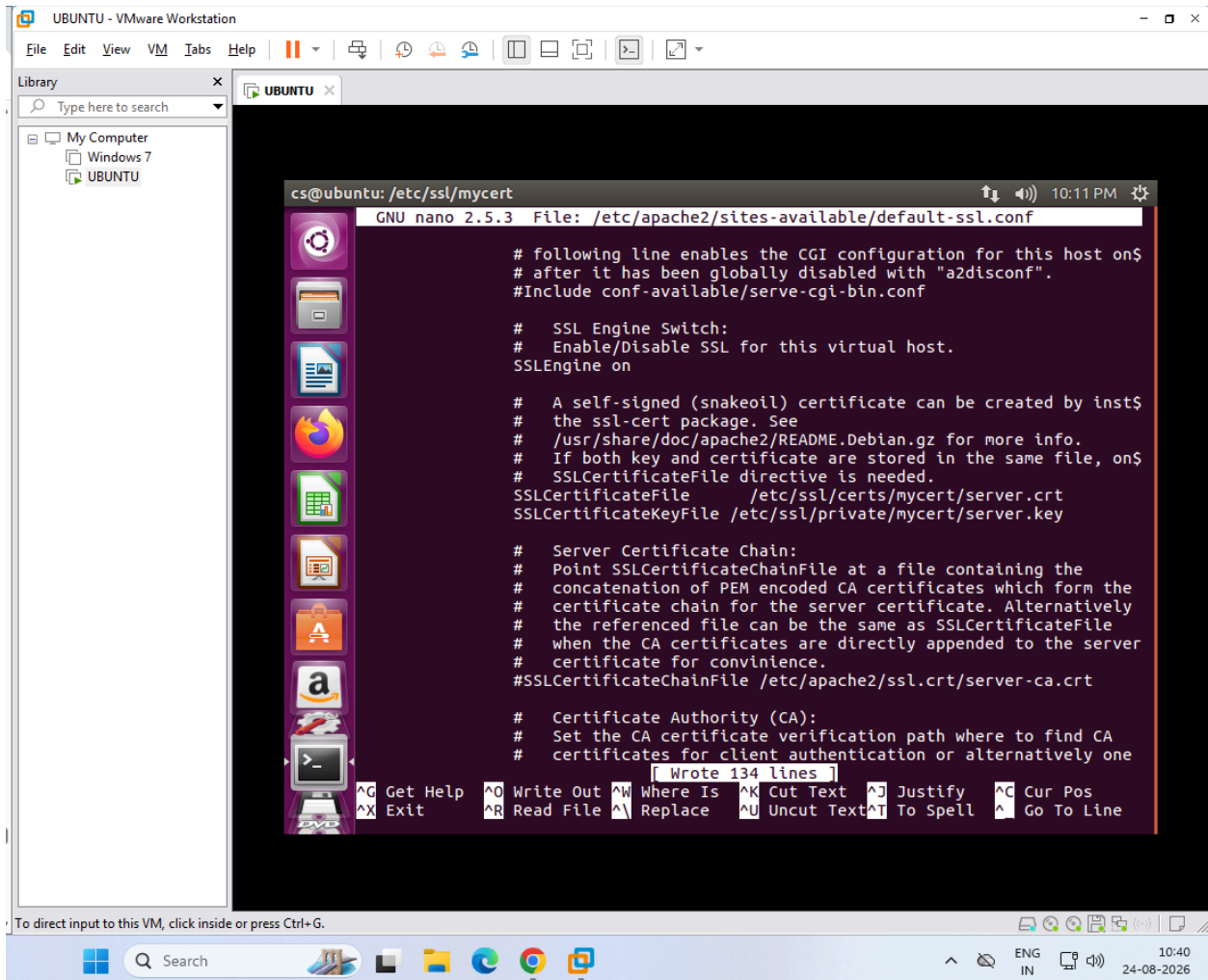
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```
sudo a2ensite default-ssl
```

Edit the SSL configuration:

```
sudo nano /etc/apache2/sites-available/default-ssl.conf
```



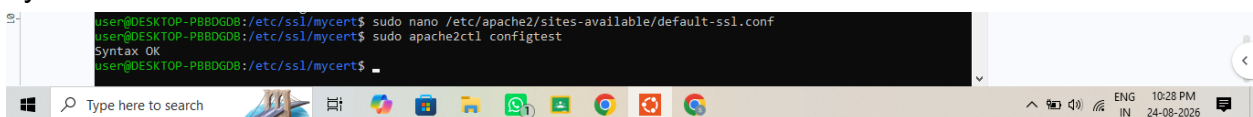
Step 9: Test Apache Configuration

Run:

```
sudo apache2ctl configtest
```

Expected result:

Syntax OK



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Step 10: Restart Apache

```
user@DESKTOP-PB8DGB: /etc/ssl/mycert
H00526: Syntax error on line 31 of /etc/apache2/sites-enabled/default-ssl.conf:
SSLCertificateFile: file '/etc/ssl/certs/mycert/server.crt' does not exist or is empty
user@DESKTOP-PB8DGB:/etc/ssl/mycert$ sudo sed -i 's|/etc/ssl/certs/mycert|/etc/ssl/mycert|g' /etc/apache2/sites-available/default-ssl.conf
user@DESKTOP-PB8DGB:/etc/ssl/mycert$ grep -n "SSLCertificate" /etc/apache2/sites-available/default-ssl.conf
10: # SSLCertificateFile directive is needed.
11: SSLCertificateFile /etc/ssl/mycert/server.crt
12: SSLCertificateKeyFile /etc/ssl/private/mycert/server.key
13: # Point SSLCertificateChainFile at a file containing the
14: # the referenced file can be the same as SSLCertificateFile
15: #SSLCertificateChainFile /etc/apache2/ssl.crt/server-ca.crt
user@DESKTOP-PB8DGB:/etc/ssl/mycert$ sudo apache2ctl configtest
H00526: Syntax error on line 31 of /etc/apache2/sites-enabled/default-ssl.conf:
SSLCertificateFile: file '/etc/ssl/mycert/server.crt' does not exist or is empty
user@DESKTOP-PB8DGB:/etc/ssl/mycert$ sudo sed -i 's|SSLCertificateFile.*|SSLCertificateFile /etc/ssl/mycert/server.crt|; s|SSLCertificateKeyFile.*|SSLCertificateKeyFil
/etc/ssl/mycert/server.key|' /etc/apache2/sites-available/default-ssl.conf
sed: can't read /etc/apache2/sites-available/default-ssl.conf: No such file or directory
user@DESKTOP-PB8DGB:/etc/ssl/mycert$ ls -l /etc/apache2/sites-available/
total 12
-rw-r--r-- 1 root root 1286 Mar 18 2024 000-default.conf
-rw-r--r-- 1 root root 4558 Aug 24 16:43 default-ssl.conf
user@DESKTOP-PB8DGB:/etc/ssl/mycert$ ls -l /etc/apache2/sites-enabled/
ls: cannot access '/etc/apache2/sites-enabled/': No such file or directory
user@DESKTOP-PB8DGB:/etc/ssl/mycert$ ls -l /etc/apache2/sites-enabled/
total 0
lrwxrwxrwx 1 root root 35 Aug 24 14:56 000-default.conf -> ../sites-available/000-default.conf
lrwxrwxrwx 1 root root 35 Aug 24 16:20 default-ssl.conf -> ../sites-available/default-ssl.conf
user@DESKTOP-PB8DGB:/etc/ssl/mycert$ sudo nano /etc/apache2/sites-available/default-ssl.conf
user@DESKTOP-PB8DGB:/etc/ssl/mycert$ sudo apache2ctl configtest
syntax OK
user@DESKTOP-PB8DGB:/etc/ssl/mycert$ sudo systemctl restart apache2
system has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
user@DESKTOP-PB8DGB:/etc/ssl/mycert$ sudo service apache2 restart
[ OK ]
user@DESKTOP-PB8DGB:/etc/ssl/mycert$ sudo service apache2 status
* apache2 is running
user@DESKTOP-PB8DGB:/etc/ssl/mycert$
```

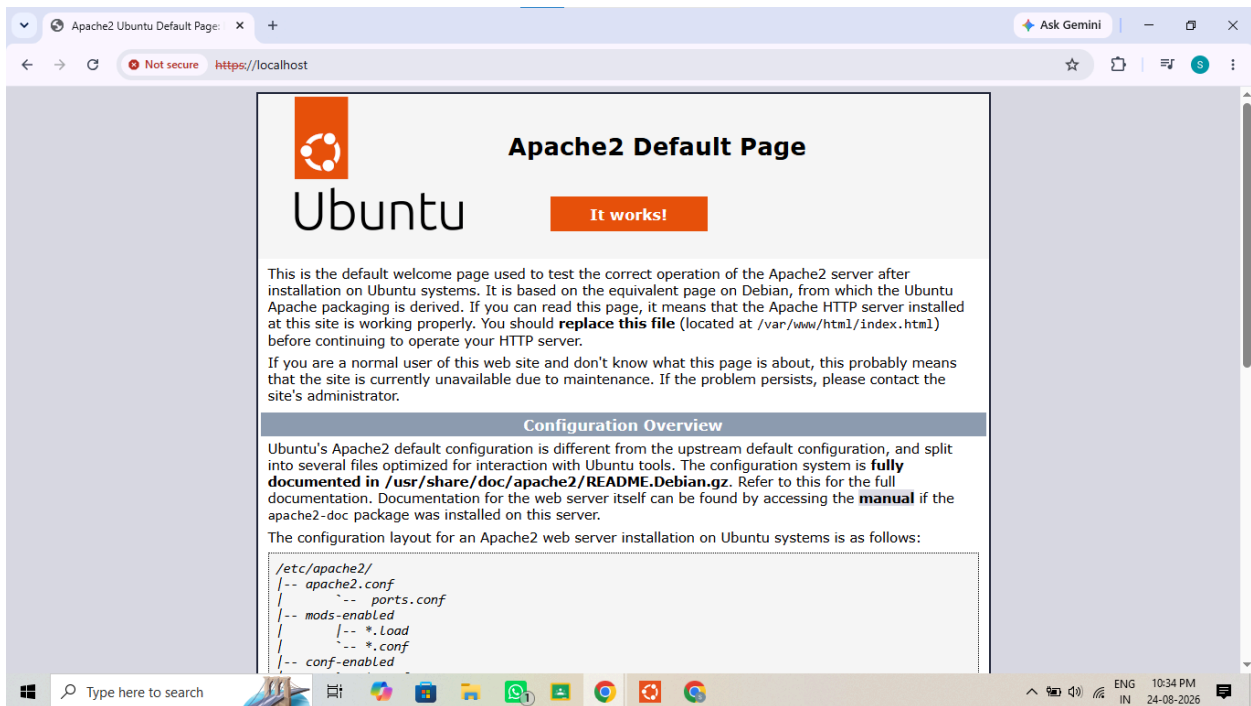
Step 11: Test Secure HTTPS Communication

Open a browser and visit:

<https://localhost>

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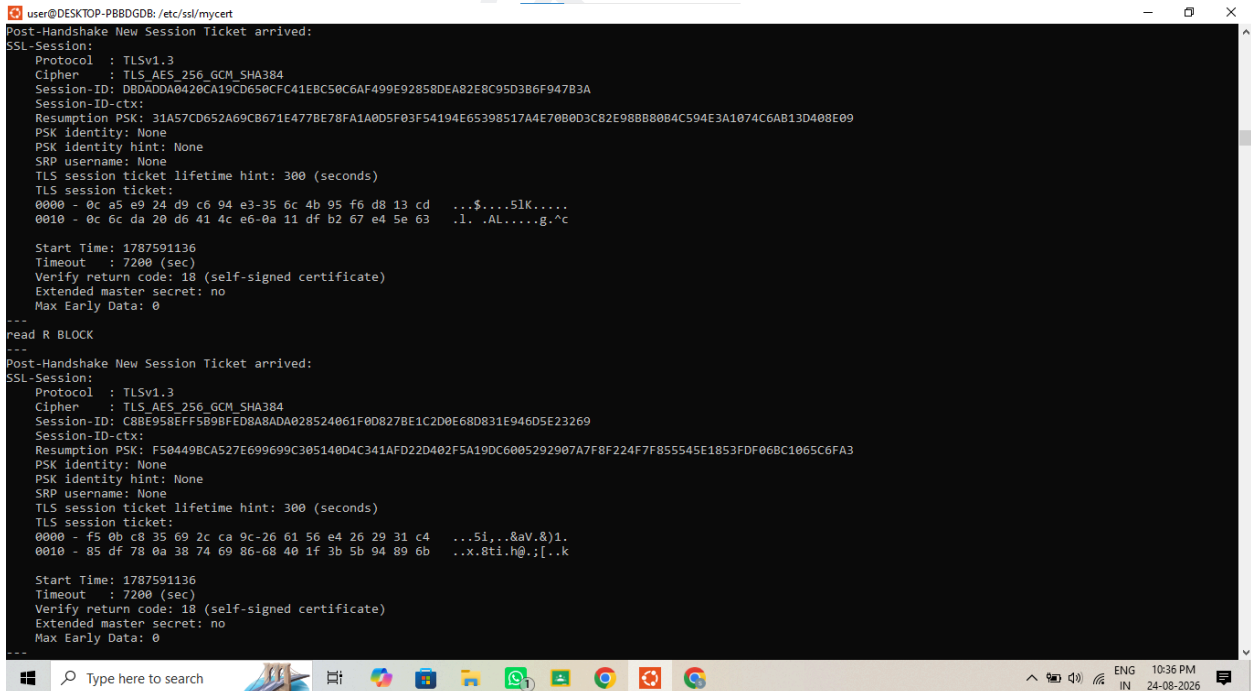
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Step 12: Verify the TLS Connection

Use OpenSSL to test the secure connection:

`openssl s_client -connect localhost:443`



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Step 13: Check the Certificate

Display certificate information:

`openssl x509 -in /etc/ssl/mycert/server.crt -text -noout`

```
user@DESKTOP-P8BDGDB: /etc/ssl/mycert
read R BLOCK
closed
user@DESKTOP-P8BDGDB:/etc/ssl/mycert$ openssl x509 -in /etc/ssl/mycert/server.crt -text -noout
Certificate:
Data:
  Version: 3 (0x2)
  Serial Number:
    50:08:fd:02:15:e7:a8:7b:4b:ba:6f:5f:86:30:8f:57:07:c3:1a:1a
  Signature Algorithm: sha256WithRSAEncryption
  Issuer: C = IN, ST = Maharashtra, L = Mumbai, O = My college, OU = CS department, CN = localhost, emailAddress = yzx@gmail.com
  Validity
    Not Before: Aug 24 16:20:03 2026 GMT
    Not After : Aug 24 16:20:03 2027 GMT
  Subject: C = IN, ST = Maharashtra, L = Mumbai, O = My college, OU = CS department, CN = localhost, emailAddress = yzx@gmail.com
  Subject Public Key Info:
    Public Key Algorithm: rsaEncryption
    Public-Key: (2048 bit)
    Modulus:
      00:9d:22:90:61:e4:de:98:27:61:2c:3c:1c:55:7c:
      15:0a:6e:d6:80:52:8b:8d:56:19:47:56:3f:a4:c6:
      a6:4d:43:ba:37:41:68:bd:8e:3b:57:6d:cf:4e:0c:
      4f:63:09:4d:0a:a4:00:8f:e8:a5:ac:f0:4d:41:a4:
      91:8b:ed:cf:65:b8:78:5e:d5:be:fd:fb:b4:56:20:
      91:c8:40:6d:ee:50:19:45:b0:45:6b:6d:04:75:b2:
      96:74:44:6c:26:9b:ae:dd:19:0c:c2:0a:22:91:3a:
      f4:16:93:17:6a:84:87:27:a6:e1:ae:da:5b:16:39:
      85:a9:bc:ea:4a:3c:0c:b7:c7:b0:14:e9:ac:fc:50:
      c9:86:6c:9e:16:44:c2:dc:4e:f4:f8:fb:cb:72:39:
      b7:fc:76:3f:32:60:a9:b8:59:a3:a7:09:f3:ed:58:
      63:f2:bb:e4:c6:25:30:d4:03:b5:38:42:fb:70:bc:
      24:3a:fd:e1:09:86:04:dc:2a:0a:7f:ee:b3:63:ce:
      b5:2f:05:c1:85:02:63:f1:9f:fa:7a:2f:45:cf:97:
      44:1b:69:ce:8b:a6:38:e5:b5:f3:b0:cd:76:1f:22:
      00:e2:a3:8a:76:19:ce:19:4f:25:af:70:87:59:d5:
      b1:e3:ac:03:90:10:5d:80:98:b6:4b:e4:5b:25:a7:
      f2:89
    Exponent: 65537 (0x10001)
  X509v3 extensions:
    X509v3 Subject Key Identifier:
      A0:0C:52:BA:E7:AC:53:C3:CC:F1:98:38:21:7D:45:C3:84:19:29:4C
    X509v3 Authority Key Identifier:
      A0:0C:52:BA:E7:AC:53:C3:CC:F1:98:38:21:7D:45:C3:84:19:29:4C
    X509v3 Basic Constraints: critical
      CA:TRUE
```

```
user@DESKTOP-P8BDGDB: /etc/ssl/mycert
c9:86:6c:9e:16:44:c2:dc:4e:f4:f8:fb:cb:72:39:
b7:fc:76:3f:32:60:a9:b8:59:a3:a7:09:f3:ed:58:
63:f2:bb:e4:c6:25:30:d4:03:b5:38:42:fb:70:bc:
24:3a:fd:e1:09:86:04:dc:2a:0a:7f:ee:b3:63:ce:
b5:2f:05:c1:85:02:63:f1:9f:fa:7a:2f:45:cf:97:
44:1b:69:ce:8b:a6:38:e5:b5:f3:b0:cd:76:1f:22:
00:e2:a3:8a:76:19:ce:19:4f:25:af:70:87:59:d5:
b1:e3:ac:03:90:10:5d:80:98:b6:4b:e4:5b:25:a7:
f2:89
Exponent: 65537 (0x10001)
X509v3 extensions:
  X509v3 Subject Key Identifier:
    A0:0C:52:BA:E7:AC:53:C3:CC:F1:98:38:21:7D:45:C3:84:19:29:4C
  X509v3 Authority Key Identifier:
    A0:0C:52:BA:E7:AC:53:C3:CC:F1:98:38:21:7D:45:C3:84:19:29:4C
  X509v3 Basic Constraints: critical
    CA:TRUE
Signature Algorithm: sha256WithRSAEncryption
Signature Value:
  95:18:c2:36:62:df:d8:51:f1:39:a7:d0:e0:17:5c:61:76:bf:
  22:41:9b:24:23:d7:32:62:9a:93:e3:24:b4:f3:95:9c:e5:53:
  1e:6c:1d:aa:1d:0c:8b:3f:24:1f:97:56:82:ce:60:a0:f6:94:
  f9:00:42:76:1b:49:f5:a2:eb:bc:12:88:90:7e:54:44:26:90:
  1b:78:81:f9:a7:f6:5d:5d:cf:54:63:9b:8b:82:f2:e0:f3:1a:
  57:ed:4c:6d:e2:9f:8e:b8:80:f8:09:16:36:ee:e8:20:fd:aa:
  9e:31:4e:8c:89:f2:27:68:4a:d5:4a:77:2b:c0:37:7a:b4:cb:
  38:e6:03:21:62:ad:7e:61:92:aa:ef:e8:81:81:1d:04:63:d5:
  3a:85:c4:06:6e:96:2f:4f:17:31:50:3c:4a:10:f0:28:4b:af:
  dd:91:02:08:ff:cc:38:ec:95:0e:b8:80:b1:bd:2e:cb:98:59:
  cd:b9:94:33:b0:a2:28:42:bb:b0:bf:08:09:91:26:35:7f:6a:
  db:30:13:6c:9d:96:df:94:cd:27:6f:bc:ad:0c:d6:bc:d7:23:
  8e:b9:d0:54:a3:d2:9c:bc:f0:a2:51:b5:6e:80:c0:f0:f0:dc:
  1c:3e:3b:7a:bc:60:45:ce:77:c0:8f:d6:dd:02:5e:55:87:60:
  17:14:fe:a5
user@DESKTOP-P8BDGDB:/etc/ssl/mycert$
```

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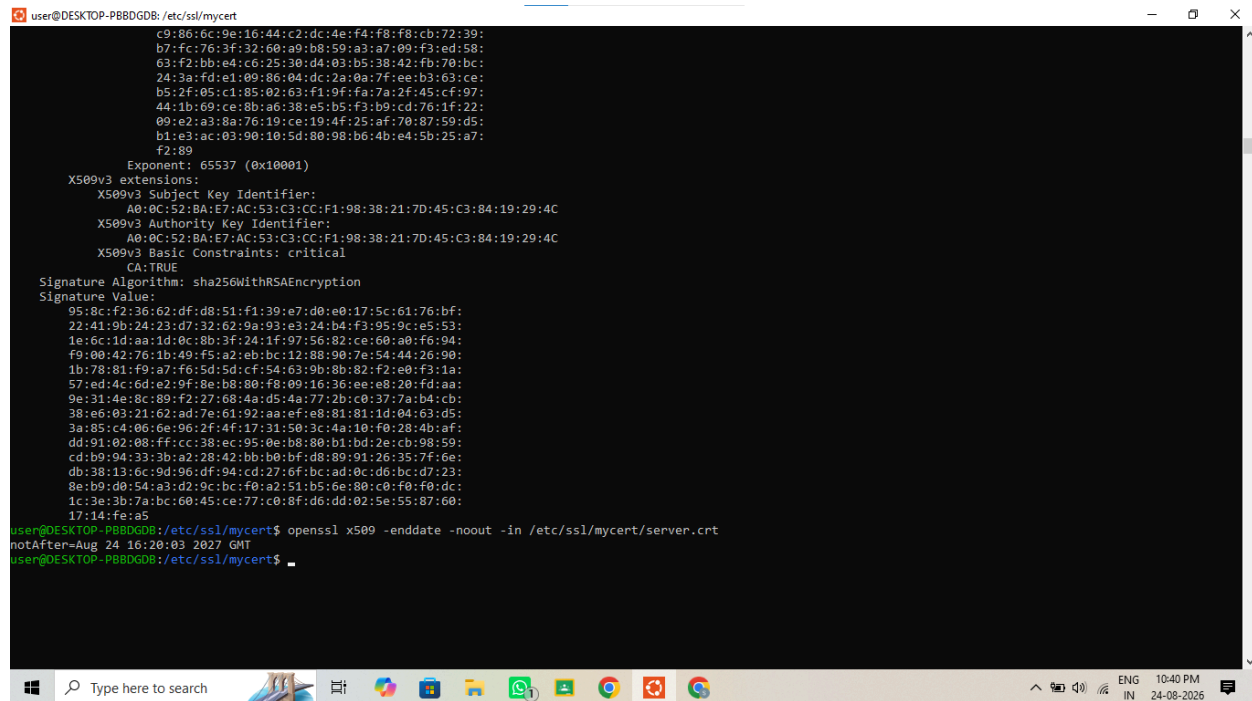
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Step 14: Secure Certificate Management

Check certificate expiry date:

```
openssl x509 -enddate -noout -in /etc/ssl/mycert/server.crt
```



```
user@DESKTOP-P8BDGDB: /etc/ssl/mycert
c9:86:6c:9e:16:44:c2:dc:4e:f4:f8:f8:cb:72:39:
b7:fc:76:3f:32:60:a9:b8:59:a3:a7:09:f3:ed:58:
63:f2:bb:e4:c6:25:30:d4:03:b5:38:42:fb:70:bc:
24:3a:fd:e1:09:86:04:dc:2a:0a:7f:ee:b3:63:ce:
b5:2f:05:c1:85:02:03:f1:9f:fa:7a:2f:45:cf:97:
44:1b:09:ce:8b:a0:38:e5:b5:f3:b9:cd:76:1f:22:
09:e2:a3:8a:76:19:ce:19:4f:25:af:70:87:59:d5:
b1:e3:ac:03:90:10:5d:80:98:b6:4b:e4:5b:25:a7:
f2:89
Exponent: 65537 (0x10001)
X509v3 extensions:
X509v3 Subject Key Identifier:
  A0:0C:52:BA:E7:AC:53:C3:CC:F1:98:38:21:7D:45:C3:84:19:29:4C
X509v3 Authority Key Identifier:
  A0:0C:52:BA:E7:AC:53:C3:CC:F1:98:38:21:7D:45:C3:84:19:29:4C
X509v3 Basic Constraints: critical
  CA:TRUE
Signature Algorithm: sha256WithRSAEncryption
Signature Value:
  95:8c:f2:36:62:df:d8:51:f1:39:e7:d0:e0:17:5c:61:76:bf:
  22:41:9b:24:23:d7:32:62:9a:93:e3:24:b4:f3:95:9c:e5:53:
  1e:6c:1d:aa:1d:0c:8b:3f:24:1f:97:56:82:ce:60:a0:f6:94:
  f9:00:42:76:1b:49:f5:a2:eb:bc:12:88:90:7e:54:44:26:90:
  1b:78:81:f9:a7:f6:5d:5d:cf:54:63:9b:8b:82:f2:e0:f3:1a:
  57:ed:4c:6d:e2:9f:8e:b8:80:f8:09:16:36:ee:e8:20:fd:aa:
  9e:31:4e:8c:89:f2:27:68:4a:d5:4a:77:2b:c0:37:7a:b4:cb:
  38:e6:03:21:62:ad:7e:61:92:aa:ef:e8:81:81:1d:04:63:d5:
  3a:85:c4:06:6e:96:2f:4f:17:31:50:3c:4a:10:f0:28:4b:af:
  dd:91:02:08:ff:cc:38:ec:95:0e:b8:80:b1:bd:2e:cb:98:59:
  cd:b9:94:33:3b:a2:28:42:bb:b0:bf:d0:89:91:26:35:7f:6a:
  db:38:13:6c:9d:96:df:94:cd:27:6f:bc:ad:0c:d6:bc:d7:23:
  8e:b9:d0:54:a3:d2:9c:bc:f0:a2:51:b5:6e:80:c0:f0:dc:
  1c:3e:3b:7a:bc:60:45:ce:77:c0:8f:d6:dd:02:5e:55:87:60:
  17:14:fe:a5
user@DESKTOP-P8BDGDB:/etc/ssl/mycert$ openssl x509 -enddate -noout -in /etc/ssl/mycert/server.crt
notAfter=Aug 24 16:20:03 2027 GMT
user@DESKTOP-P8BDGDB:/etc/ssl/mycert$
```

Step 16: Configure Strong TLS Security

Edit the SSL configuration:

```
sudo nano /etc/apache2/mods-available/ssl.conf
```


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```
user@DESKTOP-PB8DGD8: /etc/ssl/mycert
GNU nano 7.2 /etc/apache2/mods-available/ssl.conf
# Clients may prefer lower grade encryption.  You should enable this
# option if you want to enforce stronger encryption, and can afford
# the CPU cost, and did not override SSLCipherSuite in a way that puts
# insecure ciphers first.
# Default: Off
#SSLHonorCipherOrder on

# The protocols to enable.
# Available values: all, SSLv3, TLSv1, TLSv1.1, TLSv1.2
# SSL v2 is no longer supported
SSLProtocol -all +TLSv1.2 +TLSv1.3

# Allow insecure renegotiation with clients which do not yet support the
# secure renegotiation protocol. Default: Off
#SSLInsecureRenegotiation on

# Whether to forbid non-SNI clients to access name based virtual hosts.
# Default: Off
#SSLStrictSNIVHostCheck On

# Warning: Session Tickets require regular reloading of the server!
# Make sure you do this (e.g. via logrotate) before changing this setting!
SSLSessionTickets off

NG Help      NG Write Out  NG Where Is   NG Cut        NG Execute   NG Location  NG-U Undo      NG-A Set Mark  NG-I To Bracket NG-Q Previous
NX Exit      NX Read File  NX Replace   NX Paste      NX Justify   NX Go To Line NG-E Redo      NG-G Copy     NG-W Where Was NG-W Next

Type here to search
```

Restart Apache:

sudo systemctl restart apache2

```
user@DESKTOP-PB8DGD8: /etc/ssl/mycert
b7:fc:76:3f:32:60:a9:b8:59:a3:a7:09:f3:ed:58:
63:f2:bb:e4:c6:25:30:d4:03:b5:38:42:fb:70:bc:
24:3a:fd:e1:09:86:04:dc:2a:0a:7f:ee:b3:63:ce:
b5:2f:05:c1:85:02:63:f1:9f:fa:7a:2f:45:cf:97:
44:1b:69:ce:8b:a6:38:e5:b5:f3:b9:cd:76:1f:22:
09:e2:a3:8a:76:19:ce:19:4f:25:af:70:87:59:d5:
b1:e3:ac:03:90:10:5d:80:98:b0:4b:e4:5b:25:a7:
f2:99
Exponent: 65537 (0x10001)
X509v3 extensions:
X509v3 Subject Key Identifier:
  A0:0C:52:BA:E7:AC:53:C3:CC:F1:98:38:21:7D:45:C3:84:19:29:4C
X509v3 Authority Key Identifier:
  A0:0C:52:BA:E7:AC:53:C3:CC:F1:98:38:21:7D:45:C3:84:19:29:4C
X509v3 Basic Constraints: critical
  CA:TRUE
Signature Algorithm: sha256WithRSAEncryption
Signature Value:
  05:8c:f2:36:62:df:d8:51:f1:39:a7:d0:e0:17:5c:61:76:bf:
  22:41:9b:24:23:d7:32:62:0a:03:e3:24:b4:f3:95:9c:e5:53:
  1e:6c:1d:aa:1d:0c:8b:3f:24:1f:97:56:82:ce:60:a0:f6:94:
  f9:00:42:76:1b:49:f5:a2:eb:bc:12:88:90:7e:54:44:26:90:
  1b:78:81:f9:a7:f6:5d:5d:cf:54:63:9b:8b:82:f2:e0:f3:1a:
  57:ed:4c:6d:e2:9f:8e:b8:80:f8:09:16:36:ee:e8:20:fd:aa:
  9e:31:4e:8c:89:f2:27:68:4a:d5:4a:77:2b:c0:37:7a:b4:cb:
  38:e6:03:21:62:ad:7e:61:92:aa:ef:e8:81:81:1d:04:63:d5:
  3a:85:c4:06:6e:96:2f:4f:17:31:50:3c:4a:10:f0:28:4b:af:
  dd:91:02:08:ff:cc:38:ec:95:0e:b8:80:b1:bd:2e:cb:98:59:
  cd:b9:94:33:b3:a2:28:42:bb:b0:bf:d8:89:91:26:35:7f:6e:
  db:38:13:6c:9d:96:df:94:cd:27:6f:bc:ad:0c:d6:bc:d7:23:
  8e:b9:d0:54:a3:d2:9c:bc:f0:a2:51:b5:6e:80:c0:f0:f0:dc:
  1c:3e:3b:7a:bc:60:45:ce:77:c0:8f:d6:dd:02:5e:55:87:60:
  17:14:fe:a5
user@DESKTOP-PB8DGD8:/etc/ssl/mycert$ openssl x509 -enddate -noout -in /etc/ssl/mycert/server.crt
notAfter=Aug 24 16:20:03 2027 GMT
user@DESKTOP-PB8DGD8:/etc/ssl/mycert$ sudo nano /etc/apache2/mods-available/ssl.conf
user@DESKTOP-PB8DGD8:/etc/ssl/mycert$ sudo systemctl restart apache2
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
user@DESKTOP-PB8DGD8:/etc/ssl/mycert$ sudo service apache2 restart
* Restarting Apache httpd web server apache2
user@DESKTOP-PB8DGD8:/etc/ssl/mycert$ sudo service apache2 status
* apache2 is running
user@DESKTOP-PB8DGD8:/etc/ssl/mycert$
```

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T127
TYCS

MVLU

Practical 7

